

2.4.2 The Intermediate Value Theorem

Question to the Class: Suppose f is a continuous function on the set of real numbers. Suppose you know that $f(3) = -4$ and $f(10) = 2$. Must this function have a zero (i.e. an x value such that $f(x) = 0$)?

Note: The key here is to think about what the graph will look like. The next theorem addresses this question.

Theorem (The Intermediate Value Theorem):

Suppose that f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$ where $f(a) \neq f(b)$. Then, there exists a number c in (a, b) such that $f(c) = N$.

Proof: The proof is beyond the scope of this course and is not included in the text, but the picture below illustrates the idea.

Example 1: Suppose that Bob and Alice each take a 9 hour trip (during the same time interval). The temperature that Bob is experiencing at time t (in hours) is given by the continuous function $B(t)$. Similarly, the temperature for Alice is given by the continuous function $A(t)$ (both functions are continuous on $[0, 9]$). Suppose that you are told $B(0) = A(9) = -15$ and $B(9) = A(0) = 100$. Prove there is a time c such that $B(c) = A(c)$.

Example 2: Show that there is a root (i.e. solution) to the equation: $4x^3 - 6x^2 + 3x - 2 = 0$.

Example 3: Show that the graph of function $f(x) = \sin(x) - x$ obtains a y -value of -5 at least once.